AI-Powered Crop Disease & Yield Prediction Platform

Smart Agriculture Platform combining Computer Vision, Machine Learning, Time-Series Forecasting and Explainable AI

1. Project Overview

The proposed system is an AI-powered smart agriculture platform designed to help farmers monitor crop health, detect diseases at an early stage, and estimate crop yield before harvest. The platform combines two complementary AI pipelines: a computer vision model for leaf disease detection and a machine learning/time-series model for crop yield prediction. Both models are exposed through a farmer-friendly web application.

The goal is not only to provide predictions, but also to make those predictions understandable. Grad-CAM will show which parts of a leaf influenced a disease prediction, while SHAP will show which weather, soil and historical features contributed to the yield estimate. This makes the platform useful as both an applied AI system and an academic/research project.

2. What the System Does

A. Crop Disease Detection

A farmer uploads a photograph of a crop leaf. The image is preprocessed and passed to a fine-tuned deep learning model such as ResNet, EfficientNet or a Vision Transformer (ViT). The model classifies the leaf as healthy or affected by a supported disease and returns the predicted disease, confidence score and an estimated severity level.

Grad-CAM is then used to create a visual explanation that highlights the regions of the leaf that influenced the model's decision. The application can also present disease-specific recommended actions based on a trusted agricultural knowledge source rather than relying on the classifier to invent treatment instructions.

B. Crop Yield Prediction

The second AI pipeline estimates expected crop yield using agricultural and environmental information such as crop type, location, historical yield, temperature, rainfall, humidity, soil properties and other available features. A baseline model such as Random Forest or XGBoost can be compared with an LSTM or Transformer when sufficient time-series data is available.

The output can include predicted yield, an estimated risk level and the main factors influencing the prediction. SHAP provides feature-level explanations so users can understand why the model produced a particular yield estimate.

3. End-to-End User Flow

Farmer → Web App → FastAPI → Disease Model / Yield Model → Explainability → Results Dashboard

Disease flow:

Upload Leaf → Image Preprocessing → CNN/ViT → Disease + Confidence + Severity → Grad-CAM → Recommended Actions

Yield flow:

Crop/Location/Farm Data → Weather & Soil Data → Preprocessing → ML/Time-Series Model → Expected Yield + Risk → SHAP

The two results are presented together in a dashboard so the farmer can view current crop-health information and the expected yield outlook from one platform.

4. Datasets & Data Sources

Disease images: PlantVillage can be used as the initial labeled dataset because it provides a large collection of crop leaf images across many disease categories. Because many PlantVillage images are captured under controlled conditions, the project should also investigate real-world images, augmentation and background variation to evaluate generalization.

Weather: Open-Meteo or NASA POWER can provide historical and environmental weather variables. **Yield/soil:** publicly available agricultural datasets, Kaggle datasets and suitable government/open-data portals can be combined depending on the selected crops and geography.

5. AI/ML Development & Evaluation

The project will follow an experimental workflow rather than relying on a single model. For disease detection, a baseline CNN can be compared with pretrained CNN architectures and a Vision Transformer, followed by fine-tuning and data augmentation. For yield prediction, traditional regression models such as Linear Regression, Random Forest and XGBoost can be evaluated before testing LSTM/Transformer approaches where the dataset supports meaningful temporal modeling.

Disease models will be evaluated using accuracy, precision, recall, F1-score and confusion matrices. Yield models will be evaluated using MAE, RMSE and R^2. Error analysis will focus on difficult disease classes, real-world image conditions and yield predictions under unusual weather or soil conditions.

6. Explainable AI & MLOps

Grad-CAM: visualizes image regions that influenced the disease classifier. **SHAP:** explains how individual input features contribute to yield predictions. **MLflow:** tracks experiments, parameters, metrics and model versions, making model comparison reproducible.

7. Farmer-Facing Application

The application will provide a simple interface with separate disease-detection and yield-prediction workflows. The disease page will support image upload and display disease, confidence, severity, Grad-CAM visualization and recommended actions. The yield page will collect crop/farm information, retrieve weather information where possible, run the forecasting model and display predicted yield, risk and SHAP-based explanations.

8. Technology Stack

Layer	Technology
Programming	Python, TypeScript
Disease AI	PyTorch, Torchvision, timm, CNNs, ResNet/EfficientNet, Vision Transformer
Yield ML	Pandas, NumPy, Scikit-learn, XGBoost, LSTM / Transformer
Explainability	Grad-CAM, SHAP
Backend	FastAPI, REST APIs
Frontend	Next.js, React, TypeScript, Tailwind CSS, Recharts
Data / Weather	Open-Meteo, NASA POWER, public agricultural datasets
Database	Supabase PostgreSQL
Storage	Cloudflare R2 (object storage)
AI Training	Google Colab (GPU training and experimentation)
MLOps	MLflow, Git, GitHub
Deployment	Docker; Vercel; Render / Hugging Face / Cloud

9. Future Extensions

The platform is designed to remain extensible beyond the initial semester version. Future versions can add crop recommendation, irrigation and fertilizer recommendations, weather-risk alerts, additional crops and diseases, farmer history, and an agricultural RAG assistant that answers questions using verified agricultural knowledge. A later version could combine these capabilities into an AI agricultural agent that uses the disease model, yield model, weather services and knowledge base together.

10. Final Project Outcome

The final deliverable will be a deployable full-stack AI application supported by trained ML models, measurable experiments, explainability visualizations, experiment tracking and documentation. The project demonstrates practical skills across Computer Vision, Deep Learning, tabular ML, time-series forecasting, Explainable AI, APIs, frontend development, MLOps and cloud deployment.

Project Objective: Help farmers detect crop diseases early and make better production decisions by providing AI-driven crop-health insights and pre-harvest yield forecasts in one accessible platform.